

Consumer Preferences in the Used Vehicle Market: Willingness to Pay for Powertrain Type, Driving Range, Vehicle Condition, and Operating Cost

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ABSTRACT

This study employs a discrete choice experiment (DCE) to estimate consumer willingness to pay (WTP) for powertrain type and vehicle attributes in the used-vehicle market in the United States. A nationally distributed online survey was administered to 2,254 respondents who reported an intention to purchase a used car or sport utility vehicle (SUV) within two years. In the DCE, respondents were presented with repeated choice tasks in which powertrain type (conventional internal combustion engine, gas hybrid electric and battery electric), purchase price, operating cost, driving range, vehicle age, and mileage were systematically varied. Six mixed logit (MXL) models were estimated, stratified by vehicle segment (car and suv) and budget tier (low and high). The results suggest substantial heterogeneity in preferences for battery electric vehicles (BEVs), both within and across segments: mean BEV utility is estimated to be negative and highly significant throughout, but is accompanied by comparatively large standard deviations, a pattern consistent with genuine within-segment polarization rather than uniform aversion. Driving range emerges as the largest positive attribute governing BEV utility, and vehicle age is found to exert a substantial, negative, and monotonic effect on WTP across all segments considered. The only segment and budget combination for which estimated BEV WTP appears to reach statistical parity with an equivalent conventional vehicle is high-budget car buyers considering 300-mile-range vehicles. A complementary market-share simulation indicates that hybrid electric vehicles already match or exceed conventional vehicles in predicted choice share among high-budget car and SUV buyers, largely irrespective of vehicle age. These findings may carry implications for used-BEV pricing strategy, resale market development, and the design of consumer incentive programs, though they should be interpreted with the caution warranted by stated-preference data.

1. Introduction

Used vehicles are estimated to constitute approximately 70% of all vehicle transactions in the United States (Bureau of Transportation Statistics, 2023), yet the used-vehicle segment has received comparatively limited systematic attention from researchers or policymakers. This segment appears to represent a distinct buyer population, characterized by different cost constraints, vehicle-quality expectations, and information environments, a distinction taken here to warrant separate empirical treatment.

Research on electric vehicle (EV) consumer preferences has concentrated predominantly on new-vehicle buyers, and the emergence of battery electric vehicles (BEVs) in the secondary market introduces additional complexity that appears worth studying in its own right. The expanding used-BEV market may carry significance beyond its transaction volume: it is thought to extend vehicle and battery lifetimes and to support a circular economy (Guzek et al., 2024), and it may lower the effective cost of EV ownership for lower- and middle-income households (Hagman et al., 2016;

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Sheykhfard et al., 2025). A well-functioning secondary market is also understood to underpin new-vehicle residual values, which in turn affect the economics of new-EV leasing (Brückmann et al., 2021; Lim et al., 2015).

Consumer demand for used BEVs appears constrained by barriers that extend beyond those documented for the broader EV market. Used BEVs inherit range anxiety, charging-access concerns, and the upfront cost premium common to EVs generally (Sheykhfard et al., 2025), but they may face additional friction specific to the secondary market: uncertainty about prior vehicle condition and reliability is reported to be a primary purchase deterrent (Fernandes, 2023), and battery aging is understood to introduce uncertainty that age and odometer readings alone cannot fully resolve (Prieto et al., 2015).

Understanding the determinants of used-BEV purchase decisions is therefore considered important for policymakers, manufacturers, dealers, and researchers seeking to support the transition to electric mobility through the secondary vehicle market. This study reports quantitative estimates of willingness to pay (WTP) for powertrain type in the used-BEV market in the United States, disaggregated by consumer budget, vehicle type, driving range, and vehicle age. A discrete choice experiment (DCE) is used to present respondents with repeated choices between hypothetical used vehicles varying in powertrain, purchase price, operating cost, driving range, vehicle age, and mileage, allowing implicit monetary valuations for each attribute to be recovered. To the authors' knowledge, this study represents among the first attempts to estimate disaggregated WTP for powertrain type in the U.S. used-BEV market across these dimensions simultaneously.

Two dimensions are used to structure the analysis, each motivated by heterogeneity documented in the used-vehicle market. Car and SUV buyers are understood to operate in different price ranges, to have different range requirements, and to face different operating-cost profiles; pooling them risks masking preference structures that may diverge precisely on the attributes most consequential for BEV adoption. Budget tier is thought to determine which vehicles fall within a buyer's realistic consideration set and to modulate price sensitivity: lower-budget buyers in the used market appear to face fewer BEV options and to bear higher relative cost burdens (Konstantinou et al., 2025; Loh and Noland, 2024), suggesting that the price-range interaction warrants first-order attention.

This study addresses two interrelated research questions:

1. How do consumer willingness-to-pay estimates for used-vehicle attributes (powertrain type, driving range, vehicle age, mileage, and operating cost) appear to compare across vehicle segment and budget tier?
2. How do existing used BEVs appear to compete against their conventional or hybrid counterparts as vehicles age?

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on used-vehicle markets, general EV adoption determinants, and consumer preferences in the used-EV segment specifically. Section 3 describes the survey design, data collection, and sample profile. Section 4 presents the statistical methods employed. Section 5 reports the mixed logit estimates, Section 6 analyzes their implications for willingness to pay and market competitiveness, Section 7 discusses policy implications, and Section 8 is the conclusion.

2. Literature Review

The review that follows draws on three overlapping bodies of work: the broader used-vehicle market, general EV adoption determinants, and the emerging literature specific to used-EV consumer preferences.

2.1. The Broader Used-Vehicle Market

Affordability, vehicle condition, and reliability appear to be the dominant purchase criteria for used-vehicle buyers (Consumer Reports, 2023; U.S. Environmental Protection Agency, 2023). Fuel costs are also reported to factor into vehicle choice, particularly among cost-sensitive households. Used-car purchasing is reported to be more common among larger households and renters; low-income single-vehicle renters are found to carry the highest vehicle-cost burdens relative to income (Konstantinou et al., 2025).

These burdens appear to have grown since 2020. Average used-vehicle prices are reported to have climbed relative to pre-pandemic levels and have remained elevated through 2024–2025 (Cox Automotive, 2024); in April 2025, the average transaction price was \$25,547 (Kelley Blue Book, 2025). Cox Automotive (2025) report that 74% of used-vehicle buyers evaluated both new and used options, and that 71% entered the market without a fixed vehicle preference, which may indicate that buyers are searching more broadly under price pressure. C-4 Analytics (2025) report that 43% of consumers would switch brands to obtain a lower price.

2.2. General Factors in EV Adoption

Purchase price, operating costs, driving range, recharging time, charging infrastructure, environmental benefits, and government incentives have all been found to shape EV adoption decisions (Zou et al., 2020). Range and recharging time are reported to show statistically significant effects across multiple stated-preference studies (Liao et al., 2017). A meta-regression of 33 such studies finds that consumers appear willing to accept shorter range in exchange for denser charging networks, but only once BEV range already exceeds roughly 250–300 km (Rosales-Tristancho et al., 2024). Below that threshold, additional charging infrastructure does not appear to compensate for range loss, a finding that may be particularly relevant for used EVs, where buyers could plausibly discount perceived range due to battery aging. Battery warranty and depreciation rate are also reported to independently affect purchase intent (Li et al., 2020).

2.3. Used Electric Vehicle Adoption

The U.S. used-EV market is reported to have passed 400,000 units in 2023, yet studies of consumer preferences in that market remain sparse (Moras et al., 2025). Most existing work assumes price sensitivity to be the primary driver and treats it as approximately uniform across buyers. Moras et al. (2025) instead identify three distinct segments among prospective used-EV buyers: those for whom cost appears to be a binding constraint, those apprehensive about adapting to EV technology, and those apparently drawn by a favorable cost-benefit ratio and lower depreciation risk. These segments appear to share a price orientation but to arrive at it through different reasoning.

The demographic profile of current used-EV owners is reported to resemble that of new-EV owners: disproportionately male, middle-aged, highly educated, and higher-income (2024; 2024). Full-time employment and private off-street parking are also reported to predict ownership (2024). This suggests that the used market has not yet substantially reached different or lower-income buyer segments.

Prior to purchase, used-EV buyers are reported to cite price, battery performance, and charging access as their principal concerns (Loh and Noland, 2024). Lower-income buyers appear more sensitive to both price and charging availability than higher-income buyers (Loh and Noland, 2024), and concern about infrastructure is reported to correlate with outright reluctance to buy (2024). Prior EV ownership appears to work in the opposite direction: experienced EV drivers are reported to be more comfortable purchasing pre-owned vehicles, and new EV drivers report that they would consider a used EV once range and home-charging needs are met (Mashrur and Mohamed, 2025).

Used-car and new-car buyers appear to show broadly similar EV preference structures, although used-car buyers are reported to be more sensitive to fast-charging time and to the proximity of charging infrastructure to home (Zou et al., 2020). They are also less likely to have access to lease pathways, larger purchase incentives, or home-charging setups that new-EV buyers may take for granted. On price, average used-EV prices are reported to range from \$9,000–\$37,000 depending on range and vehicle class, against roughly \$14,000 for a comparable used gasoline vehicle (Konstantinou et al., 2025), a gap that price incentives may need to help close.

2.4. Synthesis and Research Gap

Used-car buyers appear more cost-conscious than new-car buyers, and current market conditions appear to have widened that difference. Price is reported to drive used-EV interest, yet buyers in that market do not appear to be a uniform group: their motivations, income, and infrastructure access differ, as does the weight they appear to assign to battery performance, range, and charging. Battery warranty and depreciation rate are reported to shape decisions beyond sticker price alone.

Existing studies draw predominantly on convenience samples or current owners, which may limit generalizability to prospective buyers. Few examine how preference weights shift across income groups, housing types, or vehicle-ownership contexts within the used-vehicle segment specifically. This study is intended to address that gap through a discrete choice experiment that estimates willingness to pay for powertrain type, driving range, vehicle age, operating cost, and purchase price across a nationally distributed sample of prospective used-vehicle buyers, disaggregated by vehicle segment and budget tier.

3. Survey Design, Data Collection, and Sample Profile

3.1. Vehicle Choice Experiment

Eligible respondents were U.S. adults reporting an interest in purchasing a used car (sedan or hatchback) or suv (including crossovers) within two years. The core instrument is a DCE comprising six repeated choice tasks in which

respondents were asked to choose among three used-vehicle alternatives with systematically varied attributes, or a “None of the above” opt-out option in each choice.

Four separate DCE designs were constructed, one per combination of vehicle body type (car, suv) and budget tier (low, high), so that the price and vehicle attributes presented to each respondent would correspond to a plausible purchase context (see Table 1). The DCE attributes are mileage, powertrain type, electric driving range, operating cost, model year, and purchase price. The DCE included three powertrain types: conventional gas vehicles (ICEV), gas hybrid electric vehicles (HEV), and battery electric vehicles (BEV) with varying driving ranges for BEV alternatives. Prior evidence identifies age and mileage as primary determinants of used-vehicle value (Prieto et al., 2015); both are accordingly included in the design, with model year serving as the observable proxy for vehicle age.

Operating cost levels were derived from observed fuel-economy data. Vehicle energy consumption (in kWh per mile for BEVs and miles per gallon for ICEVs and HEVs) was sourced from EPA ratings for 2024 model-year vehicles, stratified by powertrain and body type. These figures were combined with the 10th–90th percentile range of state-level electricity rates (\$0.11–\$0.33 per kWh) and retail gasoline prices (\$2.66–\$4.77 per gallon) prevailing in early 2025 to produce 20 approximately equally spaced operating-cost values per powertrain–body-type combination. Costs were displayed to respondents in cents per mile alongside the miles-per-gallon equivalent of a conventional gasoline vehicle at the same cost, intended to allow direct comparison across powertrain types. Operating cost was bounded by powertrain type: ICEV profiles were limited to 9–18 cents per mile, BEV profiles to 3–12 cents per mile, and HEV profiles to 6–12 cents per mile for car alternatives and 9–18 cents per mile for SUV alternatives, reflecting plausible fuel-cost ranges for each powertrain. Choice sets were drawn at random from the restricted profile space, with alternatives balanced across powertrain types and dominated profiles excluded.

Choice profiles were generated using the cbcTools R package (Helveston, 2025). Respondents were shown definitions of all attributes prior to the DCE tasks (see Table 1), which also reports detailed attribute levels. Figure 1 provides an example of the choice-task format presented to respondents.

Prior to the DCE section, respondents were asked to report their budget for the vehicle. Respondents were additionally asked to select one vehicle image (of nine) conditional on the body type (car or SUV) they had selected, choosing the style most closely aligned with the next vehicle they reported considering. The selected image was displayed throughout all choice tasks, holding body-style appearance constant across alternatives.

Table 1
BEV choice experiment attributes and levels.

Attribute	Definition	Levels			
		Low-budget Car (≤\$20,000)	High-budget Car (>\$20,000)	Low-budget SUV (≤\$25,000)	High-budget SUV (>\$25,000)
Mileage	The total number of miles a vehicle has traveled since manufacture.	All levels: 20,000–60,000 miles (5000-mile increments).			
Powertrain	The type of powertrain the vehicle runs on.	All levels: (1) <i>Conventional</i> : Gasoline or other liquid-fuel engine, such as diesel or flex-fuel. (2) <i>Gas hybrid electric vehicle (HEV)</i> : Smaller gasoline engine + electric motor + small battery. The gasoline engine recharges the battery to improve fuel efficiency. (3) <i>Battery electric vehicle (BEV)</i> : Electric motor only. Must be plugged into an electrical outlet to be refueled.			
Electric Range	Maximum distance the BEVs can travel on a full charge. Applies to BEVs only.	50–150 miles (50-mile increments)	100–250 miles (50-mile increments)	150–250 miles (50-mile increments)	200–350 miles (50-mile increments)
Operating Cost	Cost in cents per mile driven of fueling the vehicle. The equivalent fuel efficiency in miles per gallon (MPG) of a conventional gasoline vehicle is displayed in parenthesis.	Car: ICEV: 9–18 cents/mile HEV: 6–12 cents/mile BEV: 3–12 cents/mile		SUV: ICEV: 9–18 cents/mile HEV: 9–18 cents/mile BEV: 3–12 cents/mile	
Model Year	The actual year the vehicle was made.	All levels: 2015–2023 (2-year increments).			
Purchase price	The total cost of the vehicle in dollars, including down payment, monthly payments, taxes, and fees.	\$10,000–\$20,000 (\$2,000 increments)	\$20,000–\$40,000 (\$5,000 increments)	\$15,000–\$25,000 (\$5,000 increments)	\$25,000–\$45,000 (\$5,000 increments)

Question 1 of 6

Assume you are able to purchase a vehicle that looks like the one you selected (see photo). Which of the following versions of that vehicle would you be most likely to purchase? Please compare the key features shown for each option before making your selection.



If these were your only options, which would you choose?

Option 1	Option 2	Option 3	Option 4
<u>Purchase price:</u> \$ 40,000	<u>Purchase price:</u> \$ 45,000	<u>Purchase price:</u> \$ 40,000	Even if these were my best options, I would not choose any of these vehicles
<u>Powertrain:</u>  Conventional	<u>Powertrain:</u>  Gas hybrid	<u>Powertrain:</u>  Battery electric 200 miles on a full charge	
<u>Model year:</u> 2019	<u>Model year:</u> 2023	<u>Model year:</u> 2023	
<u>Mileage:</u> 20,000	<u>Mileage:</u> 25,000	<u>Mileage:</u> 25,000	
<u>Operating cost:</u> 15 cents per mile (25.7 MPG equivalent)	<u>Operating cost:</u> 9 cents per mile (37.9 MPG equivalent)	<u>Operating cost:</u> 9 cents per mile (97.2 MPG equivalent)	

Figure 1: Example of a vehicle choice task presented to respondents. Each task offers three used vehicle alternatives with varying attributes and a “None of the above” opt-out option.

3.2. Data Collection and Sample Profile

Data were collected online between December 2025 and April 2026 via two panel providers (Dynata and Prolific), with participants compensated upon completion. Following initial processing, 2,991 records were retained from sessions with a median completion time of approximately 14 minutes. Responses were subsequently screened for attention-check failures, cross-section inconsistencies, and incoherent open-ended text, yielding a final analytic sample of 2,254 respondents.

Table 2 compares the analytic sample against U.S. population benchmarks drawn from the American Community Survey (ACS) 2020–2024 five-year estimates. The sample appears to skew younger, more female, more educated, and more likely to rent than the general adult population, a pattern broadly consistent with the online-panel recruitment method and the study’s focus on used-vehicle buyers. Income distribution approximates the national profile at middle brackets, whereas the highest income tier (\$150,000 or more) appears underrepresented relative to ACS.

Table 2

Sample profile compared with the U.S. population (ACS 2020–2024 5-year estimates).

	Sample (n)	Sample (%)	ACS (%)
<i>Age</i>			
18 - 24	329	14.7	11.7
25 - 34	675	30.2	17.5
35 - 44	574	25.7	16.9
45 - 54	344	15.4	15.6
55 - 64	190	8.5	16.2
65+	121	5.4	22.0
<i>Gender</i>			
Female	1,340	59.6	51.0
Male	861	38.3	49.0
Other	49	2.2	–
<i>Race</i>			
White alone	1,543	68.5	63.3
Black or African American alone	322	14.3	11.9
Other or multiple races	389	17.3	24.8
<i>Hispanic origin</i>			
Hispanic or Latino	269	11.9	17.4
Not Hispanic or Latino	1,985	88.1	82.6
<i>Household income</i>			
Less than \$25,000	343	15.2	14.6
\$25,000 - \$49,999	374	16.6	16.7
\$50,000 - \$74,999	545	24.2	15.5
\$75,000 - \$99,999	257	11.4	12.6
\$100,000 - \$149,999	428	19.0	17.5
\$150,000 or more	304	13.5	23.0
<i>Education</i>			
High school or less	314	14.0	36.4
Some college or associate	731	32.5	27.9
Bachelor's degree	765	34.0	21.6
Graduate or professional degree	438	19.5	14.1
<i>Housing tenure</i>			
Own	1,087	52.0	65.2
Rent	1,002	48.0	34.8
<i>Housing type</i>			
Single-family detached	1,367	60.6	62.3
Single-family attached	227	10.1	6.4
Apartment (2+ units)	574	25.5	26.0
Mobile home or other	86	3.8	5.2

Notes: Sample size and shares are unweighted and exclude item nonresponse. '–' indicates no comparable ACS category. ACS universes: population 18+ (age, gender, race, Hispanic origin), population 25+ (education), and households (income, tenure, housing type).

3.3. Household Vehicle Context and Used-BEV Purchase Interest

Apart from the DCE, respondents reported their current household vehicle holdings and several factors related to EV exposure and purchase interest. Table 3 summarizes these measures. Most respondents' households appear to own at least one vehicle: 35.2% report one vehicle and 37.9% report two, while 4.0% report owning none. Among respondents reporting a current primary household vehicle, a large majority (83.8%) report driving a conventional gasoline vehicle, with hybrids (10.2%), battery electric vehicles (4.1%), and plug-in hybrids (1.8%) comprising the remainder, a distribution consistent with the sample's focus on prospective used-vehicle buyers rather than existing EV owners. Home-charger access appears split roughly evenly, with 44.0% reporting access to a home charger and 49.1% reporting none. A comparable share of respondents report knowing a neighbor who owns an EV (39.8%) as

report not knowing one (31.9%), with 28.3% unsure. Stated likelihood of purchasing a used BEV as the next vehicle appears concentrated at the unlikely end of the scale: 34.3% report being very unlikely and 20.8% somewhat unlikely to do so, compared with 8.5% very likely and 22.8% somewhat likely, with the remaining 13.6% neutral. These measures characterize the sample's baseline exposure to and stated receptivity toward BEVs, alongside the vehicle segment and budget tier stratification used in Sections 5.1 and 6.1.

Table 3

Household vehicle context and used-BEV purchase interest.

	n	%
<i>Household vehicle count</i>		
0	91	4.0
1	793	35.2
2	854	37.9
3	292	13.0
4 or more	223	9.9
<i>Current primary vehicle fuel type</i>		
Gasoline (ICEV)	1,153	83.8
Hybrid (HEV)	140	10.2
Plug-in hybrid (PHEV)	25	1.8
Battery electric (BEV)	57	4.1
Other	1	0.1
<i>Home charger access</i>		
Yes	992	44.0
No	1,106	49.1
Not sure	154	6.8
<i>Knows a neighbor who owns an EV</i>		
Yes	896	39.8
No	718	31.9
Not sure	638	28.3
<i>Likelihood of purchasing a used BEV next</i>		
Very unlikely	773	34.3
Somewhat unlikely	469	20.8
Neutral	307	13.6
Somewhat likely	514	22.8
Very likely	191	8.5

Notes: n = 2,254 unless otherwise noted; percentages exclude item nonresponse. Current primary vehicle fuel type was asked only of respondents reporting a current household vehicle (n = 1,376); all other items were asked of the full analytic sample.

4. Method

4.1. Mixed Logit Models

Consumer choice is modeled within the random utility framework. Each respondent i is assumed to select the alternative j that maximizes latent utility U_{ij} in each choice task, where utility is specified as a linear function of observed vehicle attributes augmented by an unobserved idiosyncratic error term assumed to follow an independent Gumbel distribution.

A multinomial logit (MNL) model provides the baseline specification, but two features of the choice setting are taken to favor the mixed logit (MXL) extension. First, the MNL imposes the Independence of Irrelevant Alternatives (IIA) property, which requires that consumers redistribute among remaining options in fixed proportion to current shares whenever one alternative is removed. Given qualitatively distinct powertrains, this assumption appears implausible: a consumer who prefers a conventional ICEV would, if that option were unavailable, plausibly substitute more heavily toward an HEV than a BEV, since the two share a combustion engine and a familiar refueling experience. Second, the MNL assumes preference homogeneity across respondents, a restriction that appears inconsistent with the substantial heterogeneity in EV attitudes documented in the literature, and one that is rejected empirically here

by statistically significant standard-deviation parameters $\hat{\sigma}_k$ across virtually all attributes in all models. The MXL is understood to address both concerns by modeling each preference parameter as an independent draw from a normal mixing distribution:

$$\beta_{ik} \sim \mathcal{N}(\mu_k, \sigma_k^2) \quad (1)$$

where μ_k is the population mean valuation for attribute k and σ_k captures the estimated dispersion of that valuation across respondents. A statistically significant $\hat{\sigma}_k$ is interpreted as evidence that preferences for attribute k are genuinely heterogeneous in the population, rather than uniform. The seven parameters specified as randomly distributed are the BEV powertrain indicator, HEV powertrain indicator, BEV driving range, vehicle mileage, vehicle age, operating cost, and the alternative-specific constant for the no-choice (opt-out) option. Purchase price enters as the scale parameter λ_i in the WTP-space reparameterization rather than as a randomly distributed preference weight. The internal combustion engine vehicle (ICEV) serves as the reference powertrain category throughout.

Rather than deriving WTP post-estimation as a ratio of coefficients, the models are specified directly in WTP space by reparameterizing the utility function as:

$$U_{ij} = \lambda_i \left(\sum_k \omega_{ik} x_{ijk} - p_{ij} \right) + \varepsilon_{ij} \quad (2)$$

where $\lambda_i > 0$ is a respondent-specific price sensitivity (scale) parameter and ω_{ik} denotes respondent i 's marginal WTP for attribute k , expressed directly in dollar units. Each ω_{ik} is drawn from a normal mixing distribution $\mathcal{N}(\mu_{\omega_k}, \sigma_{\omega_k}^2)$, so that the population mean WTP for attribute k can be recovered directly from $\hat{\mu}_{\omega_k}$ without further ratio computation. Uncertainty bounds are obtained by drawing 10,000 samples from the estimated joint parameter distribution and reporting the resulting 95% simulation interval.

Six MXL models are estimated across two stratification dimensions: two primary vehicle-segment models (car and SUV) and four budget-tier subgroup models (car and SUV, each crossed with *low* and *high* budget). All models are estimated in R using the `logitr` package in willingness-to-pay space, with a panel correction applied via respondent identifiers to account for the correlation among the six repeated choice observations contributed by each respondent. The mixing integral is approximated using 5,000 Sobol quasi-random draws per respondent. To reduce sensitivity to starting values and mitigate convergence to local optima, each model is initialized from 10 independent random starting points, and the run achieving the globally lowest log-likelihood is retained. Convergence is verified via the relative and absolute function tolerance criterion (Exit Status 3) in all models.

4.2. WTP for Defined Vehicle Profiles

While WTP-space estimation yields interpretable marginal valuations for each attribute, reporting coefficients in isolation is arguably too abstract for meaningful powertrain-type comparisons. A consumer choosing between a used BEV and an equivalent ICEV is unlikely to weigh powertrain type and driving range as separate, independent attributes, since range is a property specific to the BEV that does not apply to the ICEV baseline. Reporting the BEV powertrain coefficient in isolation would correspond to WTP for a BEV with zero driving range, a configuration that is not economically meaningful. To produce more readily interpretable figures, BEV WTP is therefore computed for specific vehicle configurations by combining the powertrain, range, and age parameters:

$$\text{WTP}_{\text{BEV}}(r, a) = \hat{\mu}_{\omega_{\text{BEV}}} + \hat{\mu}_{\omega_{\text{range}}} \cdot r + \hat{\mu}_{\omega_{\text{age}}} \cdot a \quad (3)$$

where r is driving range in hundreds of miles and a is vehicle age in years. This quantity is interpreted as the dollar premium (positive) or required discount (negative) for a consumer to be indifferent between a BEV with range r and age a and an otherwise identical ICEV. Sections 6.1 and 6.2 present this combined WTP at fixed combinations of range (100, 200, and 300 miles) and vehicle age. Uncertainty is propagated by drawing 10,000 samples from the estimated joint parameter distribution and reporting 95% simulation intervals.

1 **5. Results**

2 **5.1. Mixed Logit Estimates: Preference Heterogeneity by Subgroup**

3 Table 4 reports the full MXL parameter estimates for the six models, stratified by vehicle segment (car, suv) and by
4 budget tier within segment (car low, car high, suv low, suv high). For each attribute, the population mean coefficient ($\hat{\mu}$)
5 and the estimated standard deviation of the mixing distribution ($|\hat{\sigma}|$) are reported, both expressed in \$1,000 WTP-space
6 units. A statistically significant $|\hat{\sigma}|$ is taken as evidence of genuine preference heterogeneity for that attribute within
7 the subgroup. Model fit statistics are reported at the base of the table.

Table 4

Mixed logit model estimates in WTP space: by vehicle type and budget subgroups.

	By Vehicle Type		By Budget			
	Car	SUV	Car (Low)	Car (High)	SUV (Low)	SUV (High)
Mean parameters						
λ (scale)	0.173 (0.006)***	0.136 (0.004)***	0.364 (0.013)***	0.122 (0.006)***	0.245 (0.012)***	0.121 (0.005)***
BEV powertrain	-17.359 (1.209)***	-26.105 (1.935)***	-10.375 (0.820)***	-18.325 (2.681)***	-20.485 (1.982)***	-22.304 (2.906)***
HEV powertrain	0.603 (0.473)	0.805 (0.553)	-0.577 (0.322)	4.240 (1.005)***	-0.731 (0.502)	2.271 (0.811)**
Electric range	4.353 (0.642)***	4.708 (0.621)***	2.218 (0.568)***	5.234 (1.190)***	4.100 (0.784)***	3.536 (0.924)***
Mileage	-2.019 (0.122)***	-3.015 (0.165)***	-1.003 (0.078)***	-3.136 (0.272)***	-1.484 (0.139)***	-3.848 (0.248)***
Vehicle age	-0.826 (0.069)***	-1.450 (0.094)***	-0.279 (0.043)***	-1.919 (0.175)***	-0.436 (0.074)***	-2.174 (0.150)***
Operating cost	-0.746 (0.051)***	-0.861 (0.059)***	-0.412 (0.033)***	-1.054 (0.114)***	-0.581 (0.053)***	-0.900 (0.085)***
No choice	-57.452 (1.436)***	-74.663 (1.736)***	-33.136 (0.851)***	-77.831 (3.143)***	-46.547 (1.523)***	-84.371 (2.442)***
Standard deviation						
$ \hat{\sigma} $: BEV powertrain	17.017 (0.989)***	22.872 (1.172)***	9.498 (0.657)***	21.769 (2.119)***	14.543 (1.135)***	22.269 (2.211)***
$ \hat{\sigma} $: HEV powertrain	9.846 (0.546)***	12.386 (0.685)***	5.518 (0.339)***	11.557 (1.194)***	7.254 (0.594)***	14.049 (1.006)***
$ \hat{\sigma} $: Electric range	4.352 (0.907)***	0.924 (1.963)	1.897 (1.312)	6.534 (1.369)***	0.416 (1.632)	3.803 (1.255)**
$ \hat{\sigma} $: Mileage	1.381 (0.174)***	2.027 (0.214)***	0.770 (0.103)***	1.683 (0.425)***	1.091 (0.183)***	2.292 (0.302)***
$ \hat{\sigma} $: Vehicle age	0.958 (0.103)***	1.300 (0.132)***	0.430 (0.066)***	1.690 (0.215)***	0.469 (0.137)***	1.772 (0.180)***
$ \hat{\sigma} $: Operating cost	0.550 (0.066)***	0.745 (0.069)***	0.308 (0.037)***	0.693 (0.149)***	0.425 (0.061)***	0.851 (0.092)***
$ \hat{\sigma} $: No choice	13.665 (1.009)***	15.965 (1.212)***	5.152 (0.704)***	17.317 (2.153)***	8.669 (1.157)***	15.539 (1.775)***
Respondents	1,099	1,155	657	442	441	714
Choice observations	6,594	6,930	3,942	2,652	2,646	4,284
Log-Likelihood	-6,765.38	-7,198.68	-3,758.26	-2,726.13	-2,621.49	-4,448.01
Null Log-Likelihood	-9,141.23	-9,607.02	-5,464.77	-3,676.45	-3,668.13	-5,938.89
AIC	13,560.76	14,427.37	7,546.52	5,482.26	5,272.97	8,926.02
BIC	13,662.66	14,530.02	7,640.71	5,570.50	5,361.18	9,021.46
McFadden R^2	0.2599	0.2507	0.3123	0.2585	0.2853	0.2510
Adj. McFadden R^2	0.2583	0.2491	0.3095	0.2544	0.2812	0.2485

Notes: mean and standard-deviation coefficients in \$1,000 units, denominated per natural attribute unit: powertrain indicators and the no-choice constant per 1 (dummy); electric range per 100 miles; mileage per 10,000 miles; vehicle age per 1 year; operating cost per 1 cent/mile. The scale parameter λ is not a dollar-denominated WTP coefficient. Significance codes: $p < 0.001$ '***' $p < 0.01$ '**' $p < 0.05$ '*' $p < 0.1$ '.' Standard errors in parentheses.

The sign and significance pattern of the mean coefficients appears remarkably consistent across all six specifications. BEV powertrain utility is negative and highly significant in every specification ($\hat{\mu} = -10.375$ to -26.105 , $p < 0.001$ throughout), more negative for SUV buyers than car buyers at every budget tier, and more negative for high-budget than low-budget buyers within the car segment ($\hat{\mu} = -18.325$ vs. -10.375); the budget gradient appears comparatively muted within the SUV segment ($\hat{\mu} = -22.304$ vs. -20.485). HEV powertrains appear to follow a different pattern: statistically indistinguishable from the conventional baseline among low-budget buyers in both segments, but carrying a significant premium among high-budget buyers (car high: $\hat{\mu} = 4.240$, $p < 0.001$; SUV high: $\hat{\mu} = 2.271$, $p < 0.01$), suggesting that hybrid acceptance concentrates in the higher-budget market rather than being distributed evenly. Electric driving range is positive and significant in all six models, and mileage, vehicle age, and operating cost are negative and significant in all six, with each attribute's magnitude scaling consistently with both budget tier (larger for high-budget buyers) and vehicle segment (larger for SUV than car buyers). Model fit appears comparable across specifications (McFadden $R^2 = 0.251$ – 0.312), with the strongest fit obtained for the low-budget car model. Section 6.1 translates these coefficients, already expressed in WTP space, into the dollar-denominated valuations reported in Figure 2.

Beyond the mean effects, the standard-deviation estimates suggest that these are population-average patterns masking substantial within-subgroup heterogeneity rather than uniform preferences, a distinction the mean-only WTP figures in Section 6.1 cannot by themselves convey. BEV and HEV powertrain coefficients carry large, statistically significant standard deviations in all six models ($|\hat{\sigma}|_{\text{BEV}} = 9.498$ – 22.872 ; $|\hat{\sigma}|_{\text{HEV}} = 5.518$ – 14.049 ; $p < 0.001$ throughout), consistent with the negative mean BEV valuation coexisting with a meaningful share of respondents holding substantially more favorable views within every segment and budget tier. Mileage, age, and operating-cost heterogeneity is likewise significant in every subgroup. Driving range shows a less consistent pattern: heterogeneity is significant for car buyers overall ($|\hat{\sigma}| = 4.352$, $p < 0.001$) and for high-budget car buyers specifically ($|\hat{\sigma}| = 6.534$, $p < 0.001$), and for high-budget SUV buyers ($|\hat{\sigma}| = 3.803$, $p < 0.01$), but is not statistically distinguishable from zero for low-budget car buyers, low-budget SUV buyers, or SUV buyers overall, suggesting that range preferences may be comparatively homogeneous within those three subgroups, in contrast to the significant dispersion found for every other attribute throughout the table. Sections 6.1 and 6.2 build on these mean estimates to quantify what the coefficients may imply in dollar terms and for real-vehicle market competitiveness, respectively.

6. Analysis

6.1. Q1 – How Do Preferences for Vehicle Attributes Appear to Compare?

Figure 2 summarizes estimated willingness to pay for each vehicle attribute, disaggregated by vehicle segment (car, SUV) and budget tier (low, high budget); error bars represent 95% confidence intervals around the mean WTP estimates, and heterogeneity in preferences across individuals is not depicted. For the three BEV configurations (100, 200, and 300 miles of range), solid error bars denote WTP calculated at range levels falling within the range shown to that segment in the choice experiment, whereas dotted error bars denote WTP extrapolated to range levels outside that segment's design space and are accordingly interpreted with greater caution. The figure axis is scaled in thousands of dollars for readability; dollar figures below are reported at full precision. For powertrain type, high-budget buyers appear to place a statistically significant premium on HEVs relative to the conventional gasoline baseline: approximately \$4,200 among car buyers and \$2,300 among SUV buyers. Low-budget buyers, by contrast, show no significant HEV premium in either segment, with the car-segment estimate only marginally negative (approximately $-\$600$). BEV powertrains carry a WTP penalty relative to conventional vehicles in every segment and budget combination considered, though this penalty appears to narrow sharply as driving range increases. At 100 miles of range, WTP for a BEV is approximately $-\$13,100$ among high-budget car buyers, $-\$8,200$ among low-budget car buyers, $-\$18,800$ among high-budget SUV buyers, and $-\$16,400$ among low-budget SUV buyers. Extending range to 300 miles closes most of this gap: WTP rises to approximately $-\$2,600$ (high-budget car, with a confidence interval spanning zero), $-\$3,700$ (low-budget car), $-\$11,700$ (high-budget SUV), and $-\$8,200$ (low-budget SUV). The 300-mile car result for high-budget buyers is the only combination in the figure for which BEV WTP is not statistically distinguishable from the conventional baseline, which may identify added range as a particularly effective lever for closing the BEV valuation gap among higher-budget car buyers.

Vehicle age, mileage, and operating cost are each valued negatively across all four subgroups, and each penalty appears to scale consistently with budget tier: high-budget buyers exhibit larger dollar-denominated penalties than low-budget buyers for the same attribute in both segments. Mileage appears to carry the largest of the three penalties

throughout, ranging from approximately -\$1,000 (low-budget car) to -\$3,100 (high-budget car), and from -\$1,500 (low-budget SUV) to -\$3,800 (high-budget SUV). Vehicle age WTP ranges from approximately -\$300 (low-budget car) to -\$1,900 (high-budget car), and from -\$400 (low-budget SUV) to -\$2,200 (high-budget SUV). Operating cost appears smaller than both throughout, ranging from approximately -\$400 (low-budget car) to -\$1,100 (high-budget car), and from -\$600 (low-budget SUV) to -\$900 (high-budget SUV), and is overtaken by the age penalty among high-budget buyers in both segments. This pattern – larger dollar penalties among higher-budget buyers – appears to mirror the overall scale of WTP in each subgroup rather than reflecting a qualitatively different preference structure.

Comparing across vehicle segments, SUV buyers exhibit systematically larger BEV, mileage, and age penalties than car buyers at the same budget tier; operating-cost penalties appear to follow the same pattern among low-budget buyers, but reverse among high-budget buyers, where the car penalty is larger (-\$1,100 vs. -\$900). HEV premiums appear smaller for SUV buyers than for car buyers among high-budget respondents. Together, these patterns suggest that BEV valuation gaps are narrowest, and closest to parity with conventional vehicles, among higher-budget car buyers purchasing longer-range vehicles, a reading consistent with the preference structure reported in Table 4.

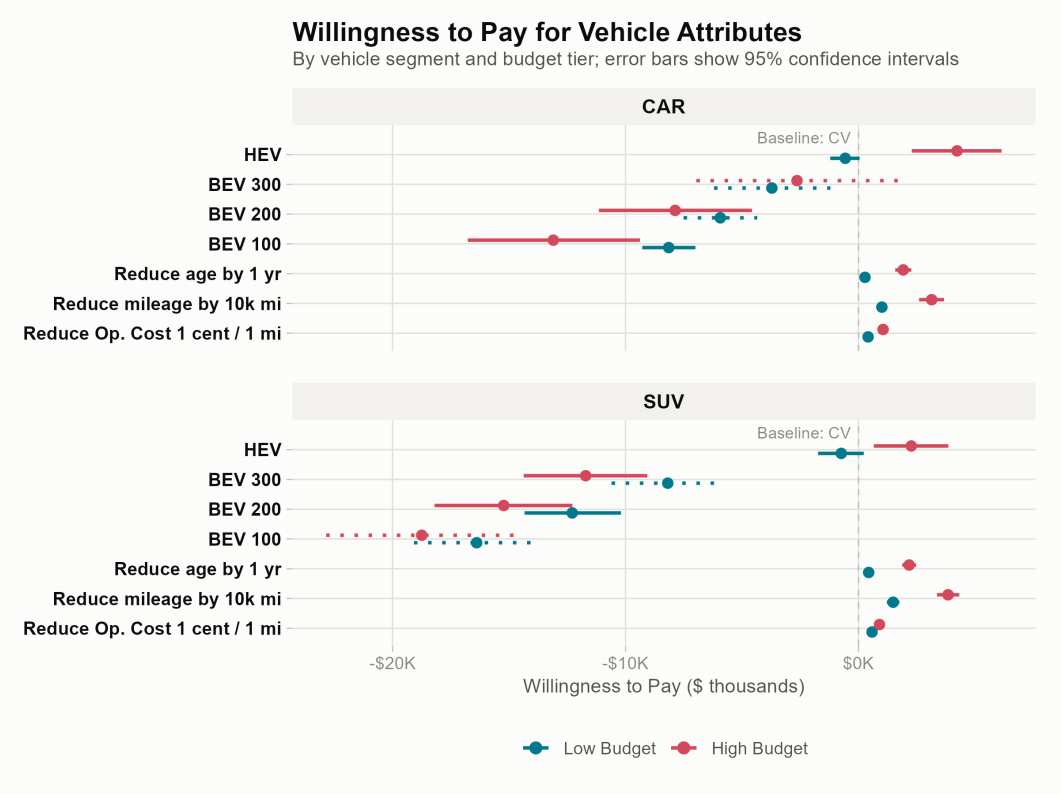


Figure 2: Willingness to pay for vehicle attributes, by vehicle segment and budget tier (95% confidence intervals shown; values in thousands of dollars). For the BEV range configurations, solid intervals mark WTP calculated within the range shown to that segment and dotted intervals mark WTP extrapolated beyond it.

6.2. Q2 – How Do Existing Used BEVs Appear to Compete Against Their Conventional Counterparts?

Figure 4 extends the willingness-to-pay estimates in Section 6.1 to a direct market-share simulation, following the approach of Helveston et al. (2014): rather than comparing isolated attribute coefficients, the estimated MXL models are used to predict a two-alternative choice probability between a real BEV and its closest currently marketed conventional or hybrid counterpart. Five matched pairs of real, currently marketed vehicles were selected for comparable body style and market positioning – three in the car segment and two in the SUV segment, spanning both BEV-vs-CV and HEV-vs-CV comparisons. For each vehicle, real used-market listings (model years 2017–2024, used inventory only) were used to fit a hedonic depreciation regression of log price on mileage and vehicle age; the fitted regression generates a

1 predicted market price at each vehicle age from 2 to 8 years, with mileage held at 10,000 miles per year of age. BEV
2 driving range is depreciated at an assumed 5% annual rate, and operating cost is fixed at the powertrain-typical value
3 used elsewhere in the design (3, 6, and 12 cents per mile for BEV, HEV/PHEV, and CV alternatives, respectively).
4 These age-specific attribute profiles are passed to the vehicle-segment- and budget-tier-matched MXL model from
5 Table 4, using the low- or high-budget model corresponding to each pair's DCE design tier ($\leq \$20,000$ or $> \$20,000$),
6 to predict the probability that a respondent chooses the BEV or HEV over its real-world counterpart at each vehicle age.
7 Rather than naming specific makes and models, Figure 4 labels each pair generically by vehicle segment, budget tier,
8 and the powertrain comparison it represents (BEV vs. CV or HEV vs. CV). Uncertainty in the predicted probabilities
9 is propagated by drawing 10,000 samples from each model's estimated joint parameter distribution; the shaded band
10 around each curve is the resulting 95% simulation interval.

11 Two of the five pairings appear to exceed parity with their conventional counterpart across the entire observed
12 age range: the high-budget car HEV-versus-CV pairing holds a substantial majority from age 2 (60% predicted share)
13 through age 8 (60%), and the high-budget SUV HEV-versus-CV pairing appears to hold a majority throughout as well
14 (54% at age 2, rising marginally to 55% by age 8). None of the three BEV-versus-CV pairings approach parity at any
15 observed age: the high-budget car pairing holds roughly a quarter share throughout (26% at age 2 and at age 5, the
16 range over which real-market listings for this pairing are available), the low-budget car pairing rises only slightly from
17 17% at age 2 to 18% at age 8, and the low-budget SUV pairing rises from 10% at age 2 to 11% at age 6.

18 Across all five pairings, predicted market share changes only marginally (at most two percentage points) over the
19 observed age range, a considerably flatter relationship than the age penalty estimated in Table 4 alone might suggest.
20 A plausible reconciliation is that the two vehicles in each pair do not depreciate at the same real-market rate: the BEV
21 or HEV's comparatively faster relative price decline with age (Figure 3) may roughly offset the shared age penalty
22 estimated in the MXL models, holding each pairing's relative competitiveness close to constant rather than improving
23 it with age. This appears broadly consistent with the logic in Helveston et al. (2014): competitiveness is understood
24 to be set by the full attribute bundle a specific vehicle offers relative to its counterpart, not by powertrain type or age
25 considered in isolation.

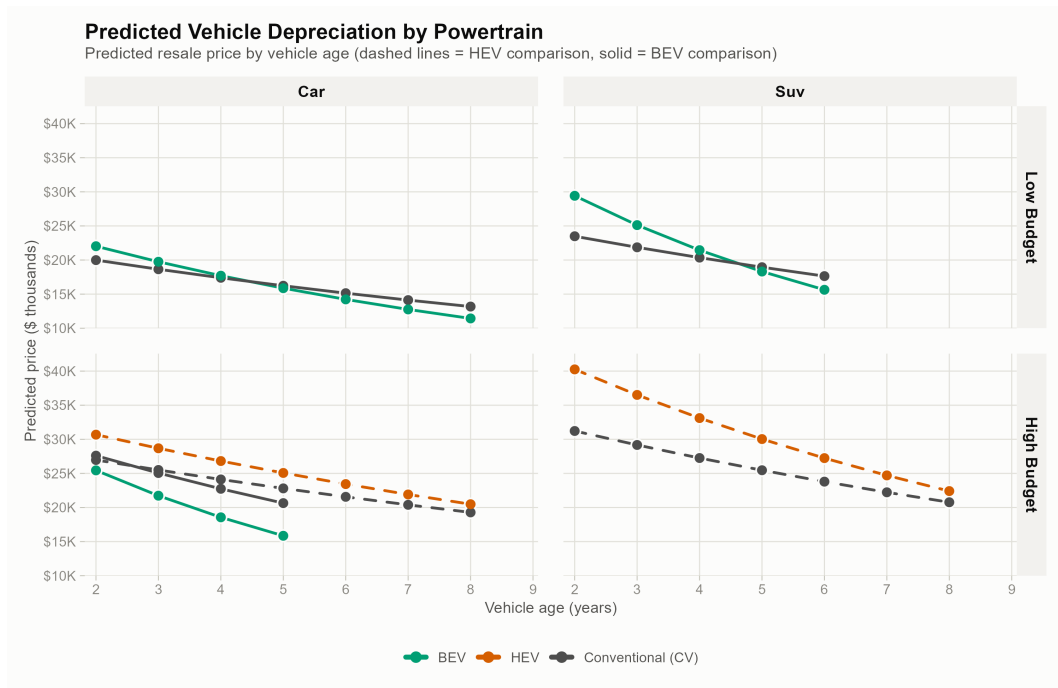


Figure 3: Predicted used-market resale price by vehicle age and powertrain for the matched vehicle pairs, from hedonic regressions of log price on mileage and vehicle age (solid lines: BEV-versus-conventional pairs; dashed lines: HEV-versus-conventional pairs)

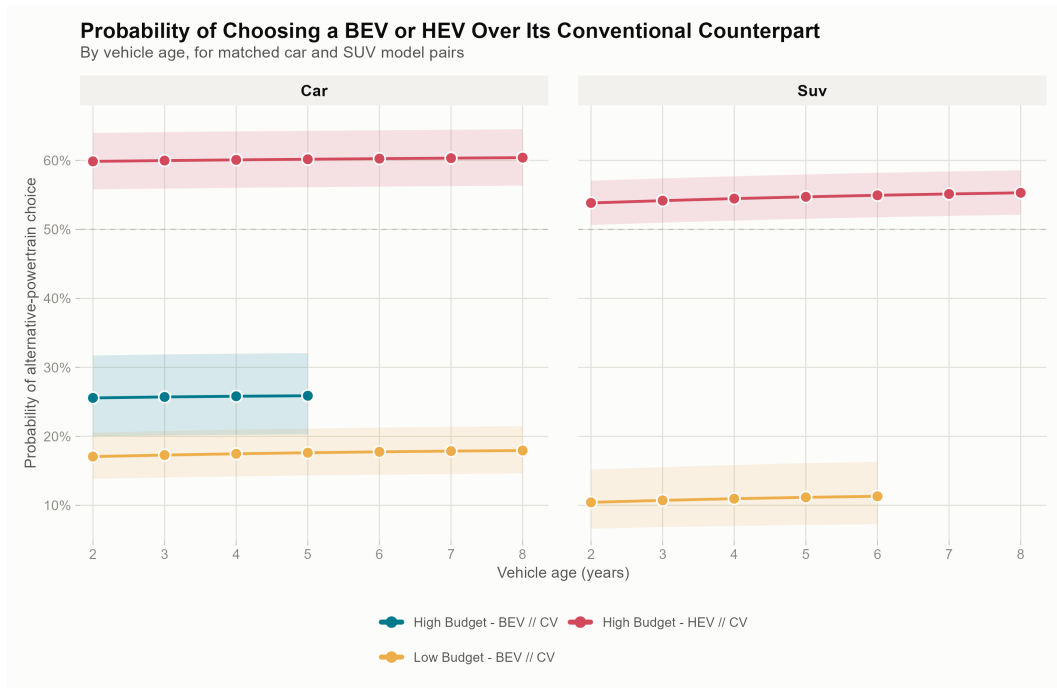


Figure 4: Probability of choosing a used BEV or HEV over its matched conventional counterpart, by vehicle age, vehicle segment, and budget tier. Lines are mean predicted two-alternative choice probabilities and shaded bands are 95% simulation intervals from 10,000 draws of the estimated parameter distribution; the dashed horizontal line marks choice parity (50%).

7. Discussion: Policy Interventions in Light of the Estimates

Most U.S. incentives for electric-vehicle adoption were designed around the new-vehicle market. The federal clean vehicle credit (Internal Revenue Code §30D) has long offered up to \$7,500 toward a qualifying new EV, and a used-vehicle counterpart arrived only in 2023, with the introduction of the used clean vehicle credit (§25E): the lesser of \$4,000 or 30% of the sale price, limited to vehicles priced at or below \$25,000 and to buyers below set income thresholds, and transferable to the dealer as a point-of-sale discount since 2024. State programs, from California's Clean Vehicle Rebate Project to scrappage-and-replacement schemes such as Clean Cars 4 All, follow a broadly similar structure. Read against the estimates in Section 6.1, the \$4,000 ceiling appears to matter for only a narrow part of the market. It is comparable to the residual valuation gap for high-budget car buyers considering longer-range vehicles, the segment already closest to parity, but appears far below what other segments would require: roughly \$8,000–\$13,000 to close the gap on a 100-mile used BEV in the car segment and \$16,000–\$19,000 in the SUV segment, and \$4,000–\$12,000 even for a 300-mile BEV.

The design of the credit also appears to disfavor the most responsive segment. The \$25,000 price cap excludes most used BEVs with 250–300 miles of range, the configuration for which WTP reaches conventional-vehicle parity among high-budget car buyers, while directing support toward shorter-range vehicles whose valuation gap \$4,000 cannot close; applied instead to the near-parity segment, an unconditioned subsidy would risk mostly rewarding purchases that would likely have occurred anyway. Consistent with Roberson and Helveston (2022), incentive structure appears to matter as much as magnitude: buyers are reported to value immediate rebates well above equivalent tax credits, and that premium appears largest for the lower-income, lower-budget, and used-vehicle buyers who dominate this market. The estimates also point toward non-price instruments. About \$15,000–\$20,000 of the WTP difference between a two- and an eight-year-old BEV exceeds conventional depreciation and most plausibly reflects uncertainty about battery state of health; standardized health disclosure, third-party certification, or transferable extended warranties (Li et al., 2020) could plausibly address that mechanism directly, and at lower fiscal cost than a cash transfer. Because the BEV valuation gap closes to statistical parity only for high-budget car buyers at long range, while HEV powertrains already appear to achieve market-share parity or better against conventional vehicles among high-budget car and SUV buyers regardless of vehicle age (Section 6.2), the results appear to favor segment-aware policy: point-of-sale incentives for

BEVs targeted by vehicle age and range, paired with continued support for HEV adoption where it is already market-competitive and with measures intended to close the battery-health information gap.

8. Conclusion

Driving range and vehicle age appear to be the dominant value determinants, each producing WTP effects that substantially exceed those attributable to budget tier or vehicle segment alone. The only segment and budget combination for which BEV WTP appears to reach statistical parity with the conventional baseline is high-budget car buyers selecting 300-mile-range vehicles. For all other combinations considered, used BEVs would need to be offered at a discount, with the required discount scaling in a manner broadly predictable from increasing vehicle age and declining range. HEV powertrains, by contrast, already appear to match or exceed conventional vehicles in predicted market share among high-budget car and SUV buyers regardless of vehicle age.

This study provides quantitative estimates of consumer willingness to pay for used battery electric vehicles in the United States, disaggregated by vehicle type, driving range, vehicle age, and budget tier. A discrete choice experiment administered to 2,254 prospective used-vehicle buyers yielded six mixed logit models from which preference parameters were recovered and heterogeneity across vehicle and budget segments characterized.

Three findings appear to carry direct implications for used-BEV market development. First, the used-BEV market does not appear to have reached price parity with conventional vehicles for the majority of prospective buyers: only high-budget car buyers selecting 300-mile-range vehicles demonstrate WTP at or near the ICE-equivalent price, while all other segment and budget combinations would appear to require discounts ranging from modest to over \$35,000. Second, vehicle age is found to be a first-order negative driver of WTP across all segments, with an estimated WTP differential of \$15,000 to \$20,000 between 2-year-old and 8-year-old vehicles within a given segment. This excess discount relative to standard ICE depreciation most plausibly reflects consumer uncertainty about battery state-of-health, suggesting that interventions improving state-of-health transparency (standardized health disclosures, third-party certification, or extended manufacturer warranties in the secondary market) could recover a portion of this penalty without requiring changes to the vehicle itself. Third, a complementary market-share simulation indicates that HEV powertrains already achieve parity or better against conventional vehicles among high-budget car and SUV buyers regardless of vehicle age, while no BEV-versus-conventional pairing approaches parity at any observed age, suggesting hybrid adoption as a comparatively low-friction near-term substitute for full battery-electric adoption in the used market.

These findings point toward a segmented market in which no single pricing or policy instrument appears uniformly effective. For higher-budget car buyers considering longer-range vehicles, the segment already closest to price parity, communication strategies emphasizing driving-range sufficiency and per-mile operating-cost advantages may be sufficient to convert marginal buyers without large price incentives. For the remaining segments, where the valuation gap is substantially larger, the federal used-EV tax credit (currently capped at \$4,000) appears to fall well short of the discounts required to reach purchase indifference. The most tractable near-term intervention for these buyers may be the promotion of hybrid electric vehicles, which already appear to achieve market-share parity with conventional vehicles among high-budget car and SUV buyers regardless of vehicle age. State-level programs designed with segment awareness – concentrating incentive value on high-budget car buyers at long range, where the market already appears close to parity – would plausibly achieve better marginal additionality than uniform subsidies applied irrespective of vehicle age, range, or budget tier.

Several limitations bound the scope of these findings. The analysis rests on stated rather than revealed preferences; hypothetical bias may affect WTP magnitudes, though its direction in this context remains uncertain. The sample was drawn from online research panels and may overrepresent information-engaged consumers. The MXL specification assumes continuous normal mixing distributions, which may inadequately represent discrete preference clusters; a latent class extension would test this possibility directly. Battery state-of-health was not varied in the vehicle-powertrain DCE analyzed here and is instead addressed in a companion battery DCE component of the broader study. Integrating findings from both instruments and cross-validating against revealed transaction prices from dealer records would likely strengthen the empirical basis for the policy inferences drawn above.

LIMITATIONS: - PHEV is not included in DCE, why?

- FUTURE WORK: - Many information collected in the survey was not utilized in this study. The study can further segment individuals based on other variables.

CRediT authorship contribution statement

Zain Hoda: Conceptualization, Data curation, Methodology, Software, Formal analysis, Writing - Original Draft, Writing - Review & Editing. **Xiatian Iogansen:** Conceptualization, Data curation, Methodology, Writing - Review & Editing. **Christina Gore:** Conceptualization, Data curation, Methodology, Investigation, Writing - Original draft preparation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Joshua D. Kneifel:** Conceptualization, Data curation, Methodology, Investigation, Writing - Review & Editing, Supervision, Project administration, Funding acquisition. **Sindhu Ranganath:** Data curation, Writing - Original Draft, Writing - Review & Editing. **John Helveston:** Conceptualization, Methodology, Writing - Review & Editing, Supervision.

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